

Days 1-5: Build and Configure a Small Office Network

Deliverable

Task 1 :- Assign static IP addresses to five devices (laptops or desktops) on your home or lab network, ensuring each is in the same subnet, and verify connectivity between them using the ping command.

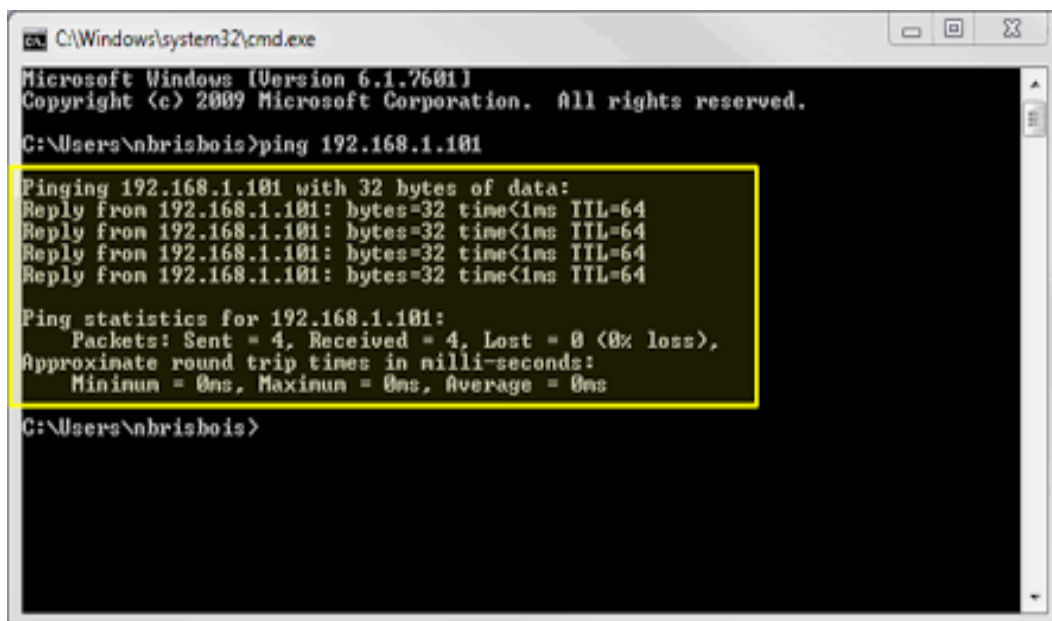
ANS :-

Static IPs (Subnet: 255.255.255.0)

- PC 1: 192.168.1.10
- PC 2: 192.168.1.11
- PC 3: 192.168.1.12
- PC 4: 192.168.1.13
- PC 5: 192.168.1.14

Steps :

1. Set IP: Enter the manual IPv4 address on each PC.
2. Ping: From PC 1, run ping 192.168.1.11 in terminal.
3. Verify: "Reply from..." means connected.



```
C:\Windows\system32\cmd.exe
Microsoft Windows [Version 6.1.7601]
Copyright (c) 2009 Microsoft Corporation. All rights reserved.

C:\Users\nbrisbois>ping 192.168.1.101

Pinging 192.168.1.101 with 32 bytes of data:
Reply from 192.168.1.101: bytes=32 time<1ms TTL=64
Reply from 192.168.1.101: bytes=32 time<1ms TTL=64
Reply from 192.168.1.101: bytes=32 time<1ms TTL=64
Reply from 192.168.1.101: bytes=32 time<1ms TTL=64

Ping statistics for 192.168.1.101:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\Users\nbrisbois>
```

Task 2 :- Set up a DHCP server (using your router or software like Packet Tracer/GNS3/VirtualBox) to automatically assign IP addresses to at least three devices, and confirm that each device receives a unique IP from the correct range.

ANS :-

1.Configure DHCP pool :

On your router or Packet Tracer Server :

- Service: On
- Start IP: 192.168.1.10
- Subnet Mask: 255.255.255.0
- Gateway: 192.168.1.1

2.Set devices to DHCP :

On PC 1, PC 2, and PC 3 :

- Go to IP Configuration and select DHCP (Automatic).

3.Verify unique Ips :

Run ipconfig on each device to confirm unique IP assignment :

- PC 1: 192.168.1.10
- PC 2: 192.168.1.11
- PC 3: 192.168.1.12

Server0

Physical Config **Services** Desktop Programming Attributes

SERVICES	
HTTP	
DHCP	
DHCPv6	
TFTP	
DNS	
SYSLOG	
AAA	
NTP	
EMAIL	
FTP	
IoT	
VM Management	
Radius EAP	

DHCP

Interface: FastEthernet0 Service: ☒ On ☐ Off

Pool Name: Pool2

Default Gateway: 192.168.10.1

DNS Server: 0.0.0.0

Start IP Addr: 192 168 10 11

Subnet Mask: 255 255 255 0

Maximum Number of Users: 255

TFTP Server: 0.0.0.0

WLC Address: 0.0.0.0

Add Save Remove

Administrator: C:\Windows\System32\cmd.exe

```
C:\windows\system32>ipconfig /renew

Windows IP Configuration

No operation can be performed on Local Area Connection 2 while it has its media disconnected.

Ethernet adapter Local Area Connection 2:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix . :

Ethernet adapter Local Area Connection:

    Connection-specific DNS Suffix . :
    Link-local IPv6 Address . . . . . : fe80::e8a1:6bb8:cab3:ebb0%11
    IPv4 Address. . . . . : 128.2.193.194
    Subnet Mask . . . . . : 255.255.254.0
    Default Gateway . . . . . : 128.2.192.1

C:\windows\system32>exit
```

Task 3 :- Configure a DNS service (using your router's DNS settings or a tool like dnsmasq) so that typing a custom hostname (e.g., musicplayer.local) in a browser on any connected device opens a specific local web page or device interface.

ANS :-

1.Add domain entry in dnsmasq :

Open the configuration file :

```
sudo nano /etc/dnsmasq.conf
```

Add this line at the bottom :

```
address=/musicplayer.local/192.168.1.50
```

2.Restart dnsmasq service :

Apply changes :

```
sudo systemctl restart dnsmasq
```

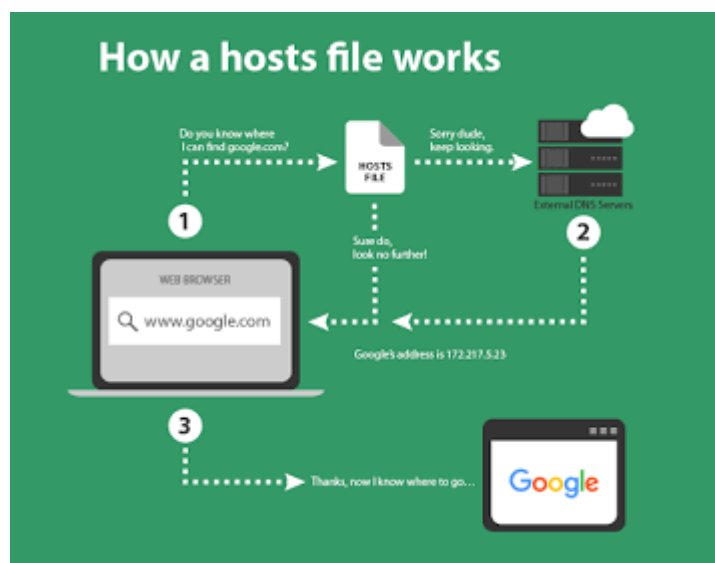
3.Test in browser :

Type http://musicplayer.local into any browser on your network. It will load your web page.

```
<dnsm
File Edit Search Options Help
#interface=wlan0
#dhcp-authoritative
domain-needed
#dhcp-range=192.168.1.100,192.168.1.150,122h
server=8.8.8.8
server =208.67.222.222
local=/home/
#dhcp-leasefile=/var/lib/misc/dnsmasq.leases
#dhcp-option=option:router,192.168.1.1
```

← upstream domain name servers

← local domain name



Task 4 :- Create two VLANs (e.g., VLAN 10: 'Workstations', VLAN 20: 'Smart Devices') on your switch or simulation tool, assign at least two devices to each VLAN, and demonstrate that devices in the same VLAN can communicate, but not with devices in the other VLAN.

ANS :-

1.Create VLANs :

Create VLAN 10 and 20 on the switch :

```
vlan 10
  name Workstations
vlan 20
  name SmartDevices
```

2.Assign Ports :

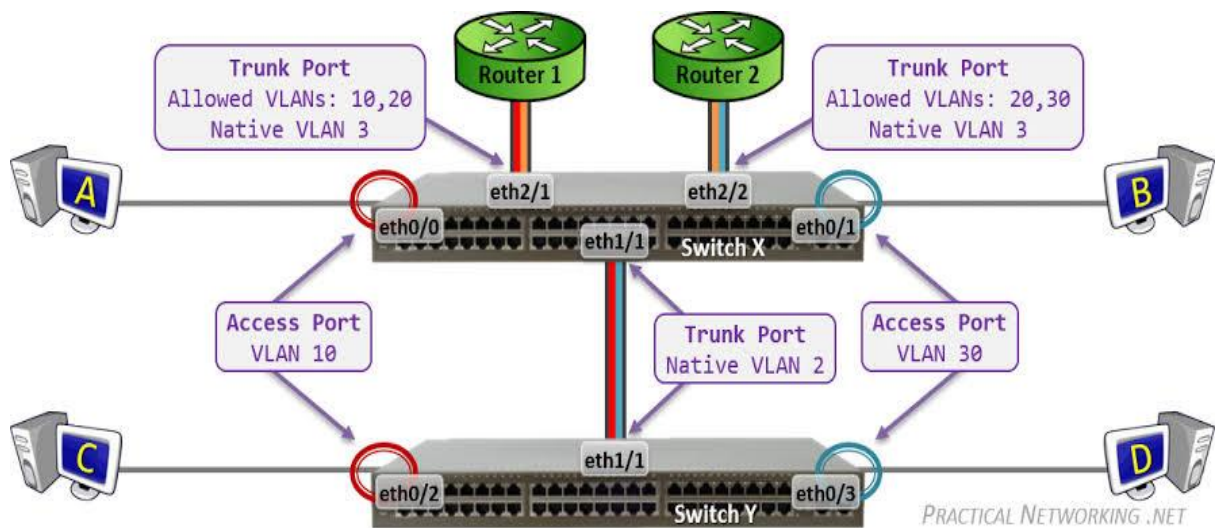
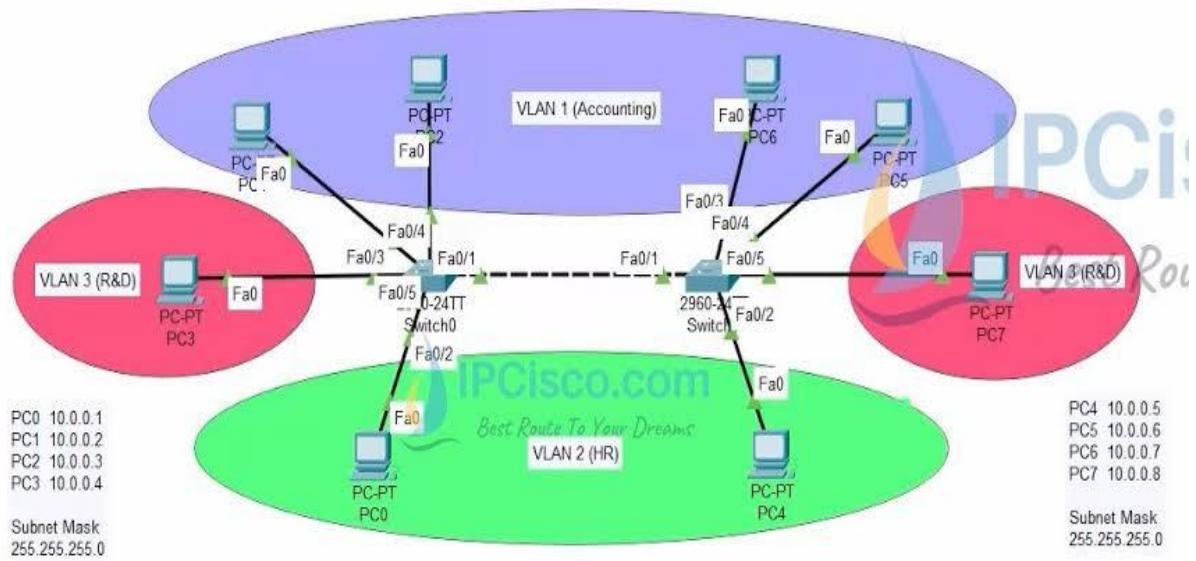
Assign ports for each group :

```
interface range Fa0/1 - 2
  switchport access vlan 10

interface range Fa0/3 - 4
  switchport access vlan 20
```

3.Verify Isolation with Ping :

- PC 1 to PC 2 (Same VLAN 10): ping 192.168.10.2 → Success (Reply from...)
- PC 1 to PC 3 (VLAN 10 to VLAN 20): ping 192.168.20.1 → Fails (Request timed out)



Task 5 :- Simulate internet access for your LAN by configuring a router with NAT (Network Address Translation), then test that devices on your LAN can access an external site (e.g., www.wikipedia.org) or a simulated external server.

ANS :-

1.Configure Router NAT Interfaces :

Set local and internet ports :

```
interface Gig0/0
ip nat inside
interface Gig0/1
ip nat outside
```

2.Enable NAT Overload (PAT):

Allow LAN devices to share the internet IP :

```
access-list 1 permit 192.168.1.0 0.0.0.255
ip nat inside source list 1 interface Gig0/1 overload
```

3.Test External Access :

From LAN PC Command Prompt :

```
ping 8.8.8.8
```

Result : Reply from 8.8.8.8 (NAT successfully translates private IP to public IP).

